

CHE 381

1. Course number and name

CHE 381: Chemical Engineering Laboratory I

2. Credits, contact hours and categorization

Credit hours: 2; Contact hours: 4 lab hours per week; Engineering

3. Instructor's or course coordinator's name

Alan D. Zdunek

4. Textbook, title, author, publisher, year

Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott, 7th Edition, McGraw-Hill, 2005.

Other supplemental materials: Provided online in the UIC Blackboard course folder to help aid in successfully preparing and carrying out the experiments. Material includes; Lab report templates, lab presentation templates, laboratory safety information, technical writing guidelines, error analysis methodology, specific articles and files related to the various experiments including vendor information on the experimental apparatus and components, theory and empirical correlations, engineering charts, and material and physical properties.

5. Specific course information

- a. Brief description of the content of the course (catalog description) Heat and momentum transfer operations associated with chemical processes. These include heat exchangers, fluid properties and fluid flow. Technical report writing, computer calculations.
- b. Prerequisites or co-requisites CHE 312.
- c. Indicate whether a required, elective, or selected elective.
Required.

6. Specific goals for the course

Provide students with a hands-on opportunity to work with physical equipment in a laboratory environment.

- a. Specific outcomes of instruction.
 - 1 Solve complex, open-ended engineering problems creatively, in a team setting.
 - 2 Integrate knowledge and skills from prior chemical engineering courses.
 - 3 Design and conduct realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations.
 - 4 Communicate effort and findings effectively through the written and oral word.
 - 5 Master and apply safety analysis and safety practices in a laboratory environment.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3) an ability to communicate effectively with a range of audiences.				X
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				X
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

Specific lab experiments performed are:

- 1) Viscosity: Non-Newtonian fluids
- 2) Distillation
- 3) Fluid Flow in Pipe Systems
- 4) Fluidized Beds
- 5) Process Control
- 6) Membrane Separation

List of Topics:

Experimental verification of theoretical principles, familiarization with engineering equipment and exposure to an engineering sense, teamwork skills; participation, leadership and team member assessment, time value; experimental planning, implementation, and completion, experimental skills; design, process safety analysis, data collection, error analysis, written and oral technical communication, integrated problem-solving.

Other topics:

Week #1: General lab safety information, technical writing and technical presentation tips and guidelines.

Week # 2: Industrial Chemical Hygiene Plan review

Week #12-14: SaChE Module: Chemical Reactivity Hazards